

Isaac Edem Adoboe

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Education

Memorial University of Newfoundland, Master of Science in Scientific Computing – St. John's, NL Sept 2025 – present

Thesis-based program focusing on physics-informed machine learning and optimization methods for computational science applications. Building on coursework-intensive Data Science foundation.

- Thesis: extreme heatwave storylines with differentiable climate models (NeuralGCM), via gradient-based optimization of atmospheric initial conditions
- Research interests: surrogate modeling, inverse problems, physics-informed neural networks

Memorial University of Newfoundland, Masters in Data Science; GPA: 4.0; Fellow of the School of Graduate Studies – St. John's, NL Sept 2024 – Aug 2025

Kwame Nkrumah University of Science and Technology, Bachelor of Science in Electrical Engineering; GPA: 3.55/4.0 – Kumasi, Ghana Sept 2017 – Sept 2021

Projects

MSc Thesis Research: Extreme Heatwave Storylines with Differentiable Climate Models Mar 2026 – present

Thesis research extending a published framework for constructing extreme heatwave storylines with differentiable climate models (Whittaker & Di Luca, 2026), using NeuralGCM and gradient-based optimization of atmospheric initial conditions.

- Developed a modular Python research package (data, model, optimization, ensemble, evaluation, and plotting modules) in JAX with a reproducible experiment pipeline over ARCO-ERA5 reanalysis data (xarray, Zarr, Google Cloud Storage).
- Ran NeuralGCM lead-time sensitivity and ensemble hit-rate experiments on the August 2025 St. John's heatwave to establish the lead time at which forecast skill for the event emerges.
- Built a multi-event heatwave atlas spanning 7 historical events across Köppen climate zones, including Moscow 2010, Pacific Northwest 2021, Japan 2018, Siberia 2020, Brazil 2023, and Sahel 2024.

Surrogate-Based Optimization for Molecular Configurations Spring 2025

Developed machine learning surrogates to replace expensive quantum chemistry calculations in protein-carbohydrate binding energy predictions, accelerating optimization

- Formulated surrogate modeling problem in 16-dimensional molecular configuration space. Trained 4 ML models (Random Forest, XGBoost, MLP, SVR) on 740 DFT-computed samples with 80/20 cross-validation and hyperparameter optimization.
- Implemented global optimization using Differential Evolution with trained surrogates as objective function approximations. Achieved speedup over direct quantum chemistry calculations while maintaining prediction accuracy for binding energy landscapes.
- Validated surrogate predictions against ground-truth DFT calculations. Analyzed model uncertainty quantification and explored multi-fidelity modeling approaches.

Fault Detection in Induction Motors using Deep Learning [BlogPost](#)

Jan - Aug 2021

Developed CNN architecture for induction motor fault classification by extracting physics-informed features from current and vibration signals via time-frequency analysis.

- Designed signal processing pipeline converting raw sensor data into physics-informed spectrograms using Continuous Wavelet Transform (CWT). Extracted time-frequency representations capturing electromechanical fault signatures from 1,500 data points across 5 fault conditions.
- Built custom CNN (4 Conv2D layers, 64 filters) achieving 61.4% multi-class accuracy on motor fault classification using TensorFlow/Keras. Implemented data augmentation strategies to address limited experimental dataset constraints.
- Analyzed model interpretability by visualizing learned feature maps and correlating them with known physical fault mechanisms (bearing degradation, winding failures, phase imbalances).

Fine-Tuned CodeGen Model [GitHub](#)

Winter 2025

Fine-tuned transformer-based CodeGen-350M on CodeSearchNet Python corpus using HuggingFace Transformers. Optimized training with GPU acceleration and evaluated using BLEU, CodeBLEU metrics for code completion quality.

- Trained on 2,700 samples (300 validation) with GPU acceleration. Implemented learning rate scheduling, warmup steps, and mixed-precision training for efficient convergence. Evaluated using BLEU and CodeBLEU metrics

Experience

Data Center Intern, Center for Analytics, Informatics, and Research (CAIR), Memorial University – St. John's, NL

Feb 2026 – present

AI and software development internship at Memorial's Research Data Centre, a research and innovation hub in partnership with IBM, building and deploying LLM-powered applications for university service teams and industry clients.

- **AI Adaptive Training Platform (Industry Client):** Architected and built the full-stack application for an AI-powered adaptive training platform: React/TypeScript SPA, Django backend, and MongoDB case repository, deployed with Docker Compose. Developed LLM-driven content generation with versioned prompts against a CAIR-hosted open-weights model, built an authoring studio with a question bank for subject-matter experts, and implemented beginner-to-advanced adaptive difficulty progression.
- **RIS Policy & Procedures Chatbot:** Built and deployed an LLM-powered chatbot for Memorial's Research Initiatives & Services (RIS) team, answering questions grounded in university policies and procedures. Designed the policy-document ingestion pipeline, A/B tested retrieval and prompt configurations, realigned model responses for accuracy, and delivered the production chat UI; demoed to university stakeholders.
- **RIS Ticketing System:** Developed an internal ticketing web application for the RIS team. Refactored it and the chatbot onto a shared application core, rebuilt the UI/UX with TailwindCSS and Web Awesome, and implemented feature flags to right-size scope to stakeholder needs.
- **Deployment & Operations:** Containerized both applications (Docker/Podman) and deployed to Linux VMs with admin panel and support-email integration. Managed LLM API key security, debugged inference failures, and ran phased testing from internal team to single external tester to full client-team rollout.
- **Collaboration & Documentation:** Ran 1-1 requirements sessions with client stakeholders, drove PR-review cycles with the development team, migrated the codebase into the CAIR organization repository, and authored user manuals and technical documentation for both applications.

Technology Portfolio Support Analyst, Newmont Mining Corporation – Akyem Mine, Ghana Nov 2021 – July 2024

Promoted from End User Support Analyst (Aug 2022) to Technology Portfolio Support Analyst. Transitioned from frontline technical support to infrastructure development and data automation projects supporting 900+ users across mining operations.

- **Data Automation & Analytics (2023-2024):** Built Python-based reporting pipeline automating Lenel security system data extraction and transformation for finance/HR stakeholders (Power BI dashboards). Replaced manual processes with scheduled delivery reducing turnaround from days to hours.
- **Infrastructure Deployment (2023-2024):** Coordinated 8+ major IT infrastructure projects including LIMS modernization, workforce management systems, and mining software deployments (MineSight, Deswik). Bridged technical requirements between vendor solutions and operational needs.
- **Technical Operations Support (2021-2022):** Provided frontline technical support achieving 90% first-contact resolution. Built foundational expertise in enterprise systems, Azure cloud infrastructure, and mining-specific software platforms.

Web Developer, Envision Digital Signage Network – Kumasi, Ghana (Remote) Feb 2021 – Dec 2022

Part-time remote web development role begun during final year of undergraduate studies, designing, developing, and supporting the company's main website and React-based applications

- **Web Application:** Developed responsive React application for digital signage management, contributing to 100% sales growth. Reduced page load times by 40% through code splitting and performance optimization.
- **Component Architecture:** Created reusable component libraries (dashboards, data visualizations, forms) using React Context API and custom hooks.
- **APIs, Backend & SEO:** Integrated third-party APIs and built backend functionality with SQL databases and JavaScript libraries. Implemented responsive design and search engine optimization to increase visibility and attract clients.

Intern, Ghana Career Program 2021 Cohort, Global Startup Ecosystem (GSE) – Remote Mar 2021 – July 2021

Career development program fostering entrepreneurship in Ghana, combining mentorship, training, and networking with hands-on project work

- Collaborated remotely on social media design projects, ensuring consistent branding and high-quality visuals aligned with GSE's target audience and brand identity.
- Managed and analyzed program data in Excel; received mentorship on startup funding, essential skills, and navigating the startup ecosystem.

Electrical and Electronic Engineering Intern, Fredmef Group Limited – Accra, Ghana Sept 2020 – Dec 2020

Engineering internship supporting building projects, from site drawings to client and logistics coordination

- Reviewed and corrected site drawings for 4 buildings using AutoCAD.
- Coordinated with clients, supply, and logistics teams to make provisions for new and existing projects; organized project information in Excel for analysis and research.

Electrical Engineering Intern, Volta River Authority – Akosombo, Ghana May 2020 – Aug 2020

Summer internship supporting high-voltage electrical distribution systems for national power infrastructure

- **High-Voltage Systems:** Assisted maintenance and troubleshooting of 161kV and 330kV transmission lines ensuring continuous power delivery. Participated in preventive maintenance scheduling and fault analysis.
- **Power Distribution Analysis:** Processed load flow data and voltage profiles for distribution optimization studies, identifying system improvements and efficiency opportunities.

Skills

Scientific Computing & Optimization: Numerical optimization (SciPy, Differential Evolution), surrogate modeling, physics-informed ML, inverse problems, differentiable climate modeling (NeuralGCM, JAX)

Machine Learning Frameworks: TensorFlow, Keras, PyTorch, scikit-learn, model uncertainty quantification

Programming & Numerical Methods: Python (NumPy, SciPy, pandas, xarray, matplotlib), MATLAB, Git/GitHub, high-performance computing (Jupyter, Compute Canada), Bash scripting

Signal Processing & Analysis: Time-frequency analysis (CWT), spectral methods, sensor data processing, multi-modal data integration

Software Engineering & Deployment: LLM/RAG application development, full-stack web development (React/TypeScript, Django, TailwindCSS), MongoDB, containerization (Docker, Podman), Linux VM deployment, Git/GitHub PR workflows

Domain Expertise: Climate modeling and atmospheric physics, Computational physics, molecular modeling, electro-mechanical systems, hybrid physics-ML methods, data-driven modeling